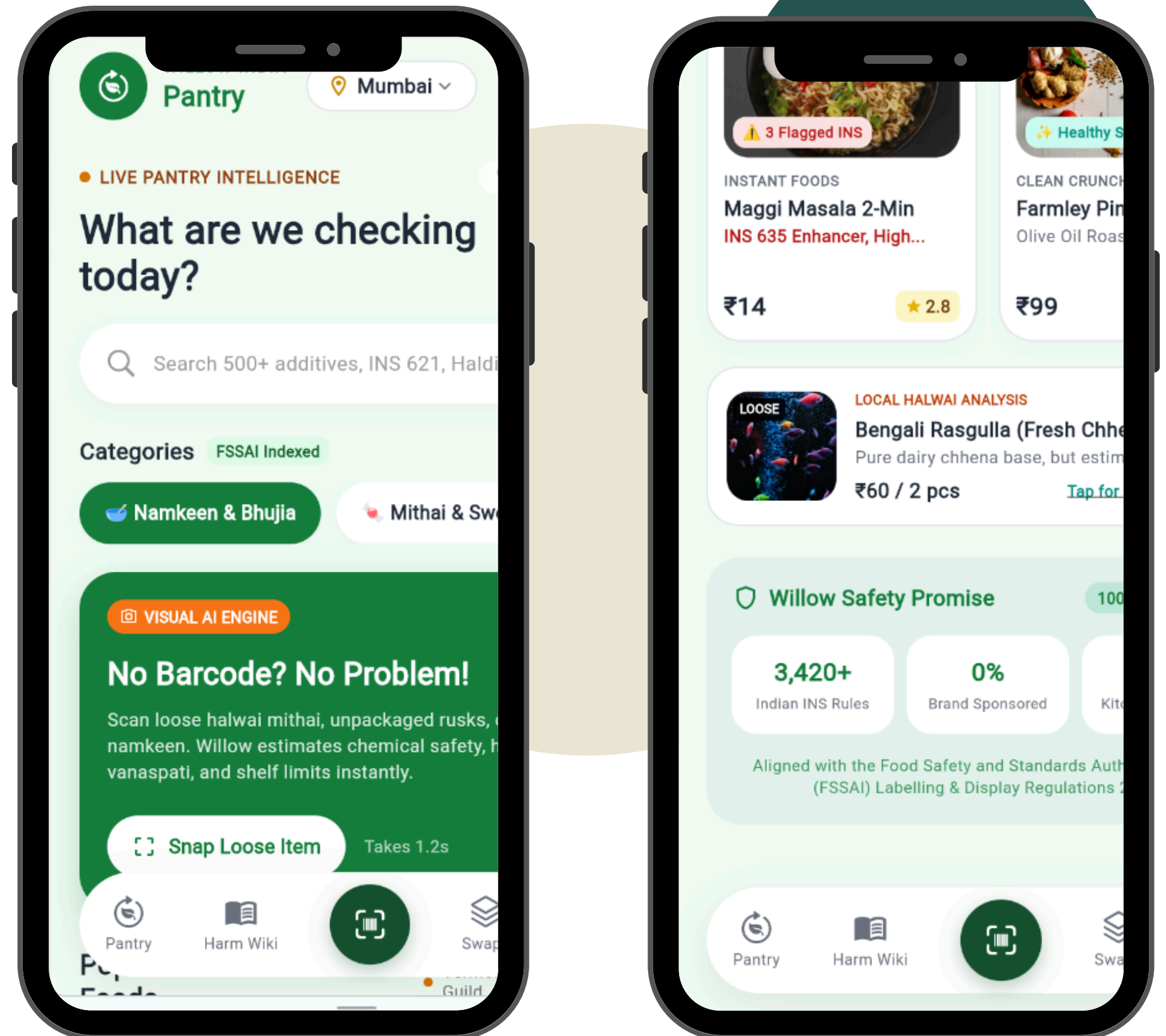


WILLOW

● PROGRESS UPDATE



ABOUT US

A user photographs a packaged product. The app reads the ingredient list directly from the photo, no barcode lookup, flags ingredients known to be harmful with plain-language health explanations, and suggests healthier alternatives in the same category and price range.



PHOTO

USER CAPTURES OR UPLOADS A PACKAGING PHOTO.

SNAP!



EXTRACT & CLASSIFY

READS THE LABEL, CHECKS INGREDIENTS AGAINST HARM DATA.

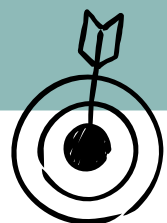
SCAN



RESULT

⚠ 2 FLAGGED INGREDIENTS
✓ ALTERNATIVES · HIGH CONFIDENCE

DONE



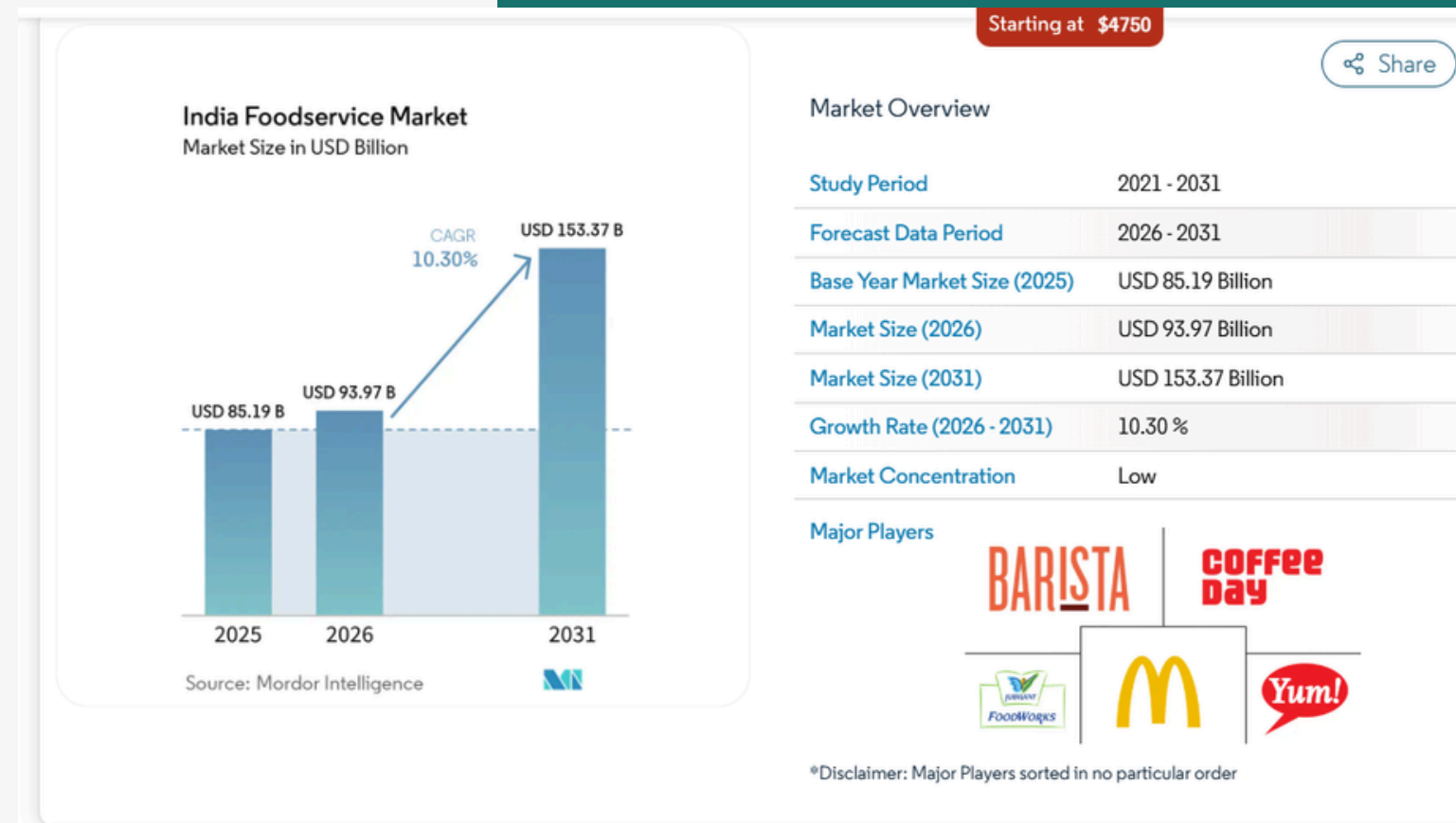
EXECUTIVE OVERVIEW & PROBLEM



MARKET GAPS & PROBLEM STATEMENT

Limitations of Existing Solutions

- **Barcode Dependency Gap:** Apps like Yuka rely on global barcode DBs, failing on local, regional, or unbranded Indian goods.
- **Missing Ingredient Panels:** Unbranded local goods (loose namkeen, sweets) often lack printed ingredient lists entirely.
- **Foreign Regulatory Scoring:** Existing platforms apply EU/US standards rather than FSSAI guidelines, ignoring Ayurvedic ingredients.
- **Food Safety & Adulteration:** Frequent spice adulterations and hidden synthetic additives mandate localized risk checking.



METHODOLOGY & ARCHITECTURE

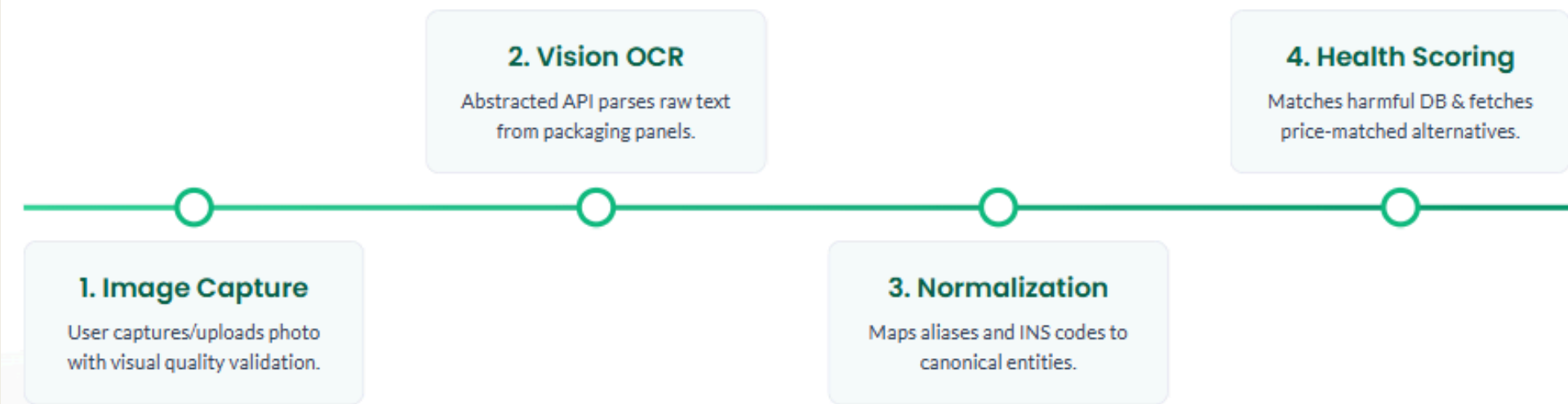
Path A: Ingredient Panel Present

Performs OCR text extraction, normalizes E-numbers/aliases, maps FSSAI risk levels, and displays exact ingredient risk scores.

Path B: Missing Ingredient Panel

Classifies product into sub-category (e.g., Namkeen, Sweets) and surfaces category-level estimated risk profiles with clear disclaimers.

End-to-End Processing Workflow



Req ID	Module	Feature Description	Priority
FR-1 & FR-4	Capture & OCR	Camera photo capture & raw text extraction via OCR service	MUST-HAVE
FR-5 & FR-6	Parsing & Alias	Text parsing into structured lists & INS/E-number normalization	MUST-HAVE
FR-8 & FR-10	Risk Analysis	Harmful DB comparison, health effect notes & safety scoring	MUST-HAVE
FR-12 & FR-13	Fallback Engine	Sub-category risk fallback estimation for unlabelled local items	MUST-HAVE
FR-16 & FR-17	Alternatives	Cleaner alternative recommendations matched by price band	MUST-HAVE

SYSTEM ARCHITECTURE & TECH STACK

• Frontend & API Layer

~Akanksha

Frontend Clients:

- Web Application: Responsive Web Dashboard built with JavaScript (React.js / Next.js).
- Mobile Application: Cross-platform Mobile App built with Flutter (iOS & Android) for native camera capture and scanning.

• Processing & Data Layer

~Aditya

- Vision Interface: Abstracted internal API interface allowing provider hot-swapping.
- Backend Framework: Modular Monolith REST API using Node.js with Python FastAPI serving both the Web and Mobile clients.

PROJECT STATUS

Where things stand today

Moving from a locked MVP scope into a working, demonstrable prototype.



Deep Research

Market validation, competitor analysis (Yuka, FactsScan), FSSAI-based differentiation strategy completed.

IN PROGRESS



Extraction Pipeline

OCR / ingredient-extraction engine built, integrated with Gemini Vision API for reading packaging text.

DONE



Flutter Frontend

Cross-platform mobile app UI in progress — capture flow and results screens underway.

DONE



Project Web Page

Public-facing web page created to present and showcase the project.

DONE

PROJECT SUCCESS & REAL-WORLD VALIDATION

Jerry | NutriGraphAPI Sep 9
to me, Santa

Hi Willow team,

I've been looking at Willow, and I like the focus on a part of nutrition where accuracy is not a nice-to-have. For people managing kidney health, the numbers behind minerals and macros can carry real weight.

That is why I reached out. Products in this area depend on data that is not only available, but clinically sensible, current, and consistent across foods. Sodium, potassium, phosphorus, and protein values are exactly the kind of details users need to trust.

The difficult part is keeping that data right at scale. Most teams end up doing too much of it themselves: scraping sources, reconciling conflicts, patching gaps, and checking new foods as they appear.

That is the specific problem NutriGraphAPI solves. NutriGraphAPI returns structured nutrient and label data for barcode-indexed packaged products, so renal-focused scanners can evaluate the fields available for each UPC/EAN without owning the packaged-product data layer.

Its scope is deliberately precise: NutriGraphAPI looks up packaged products by UPC/EAN barcode. It does not provide recipes, dishes, menu records, or standalone lookup for unbarcoded ingredient names.

higher retention, and fewer moments where a shaky or missing number undermines trust in the product.

For Willow, that can mean more dependable nutrition context without turning data maintenance into a second product.

Full technical docs and JSON structures are here:
<http://5.189.155.146:8095/r/email%3Aathakurakanksha288%40gmail.com>

The practical first step is a coverage check. Reply with 10-15 representative UPC/EAN barcodes, the target markets, likely monthly lookup volume, and any ideas you have for how this data could become useful inside your product. I am especially interested in joint experiments where better packaged-product data could unlock more frequent use, higher-value upgrades, better recommendations, or happier customers. If the live validation looks useful, I would be happy to set up a free trial API key around that exact workflow.

If there are specific data gaps you are already fighting, I would genuinely like to hear about them. That is usually the fastest way to tell whether we are a fit.

Best,
Jerry
VP, Outbound Solutions

NutriGraphAPI

- **Live Scanning Validation**

Demonstrating high-precision OCR parsing and real-time health risk evaluation on real consumer food items.

The validation section demonstrates real-world market interest and external verification for the Willow project based on the outreach email from Jerry, VP of Outbound Solutions at NutriGraphAPI.

- **Inbound Industry Recognition:** NutriGraphAPI reached out directly to the Willow team, validating the system's focus on high-precision nutrition tracking—specifically emphasizing that accurate mineral and macro data (Sodium, Potassium, Phosphorus, Protein) is critical for high-stakes use cases like renal/kidney health monitoring.
- **Validation of Core Problem:** The outreach confirms that managing nutrition data at scale is a major industry challenge where standard scraping, conflict reconciliation, and patching gaps frequently fail.
- **Potential API & Data Partnership:** NutriGraphAPI proposed a coverage test and a free trial API key to run joint experiments, exploring how integrating structured barcode nutrient data can enhance Willow's scan accuracy, boost user trust, and deliver better product recommendations without forcing the team to build and maintain an entire packaged-product data layer from scratch.
- **Live Scanning & System Verification:** Demonstrates operational accuracy in OCR parsing, health risk evaluation, and smooth fallback handling across real consumer food items.

FUTURE SCOPE

Willow ("Know what's really in it") is an AI-driven ingredient scanner tailored specifically for the Indian market. It addresses a major gap in existing food scanning apps, which rely heavily on international barcode databases and Western regulatory standards that fail for unbranded, regional, or unindexed Indian food items.

- Ecosystem Expansion: Building open API frameworks to foster seamless third-party integrations and developer partnerships.
- Sustainability Initiatives: Transitioning to green computing practices and minimizing carbon footprints across operations.
- AI & Automation Integration: Leveraging advanced predictive algorithms and automated workflows to increase operational efficiency.
- Global Scale & Expansion: Penetrating international markets and tailoring features for cross-border adaptability.
- Enhanced Security & Compliance: Implementing end-to-end data encryption, zero-trust architecture, and strict regulatory adherence.



THANK YOU

● FOR YOUR NICE ATTENTION

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